

Lecture 6 (9/25)

Variance, Normal models and Linear Regression

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Related quantities:

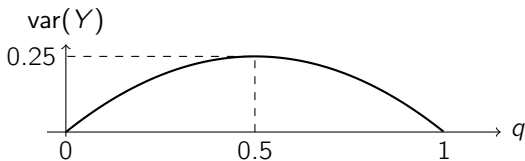
- The standard deviation is $\sigma = \sqrt{\text{var}(Y)}$
- The coefficient of variation is $\text{CV}_Y = \sigma/\mu$ where $\mu = E[Y]$.

Example

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

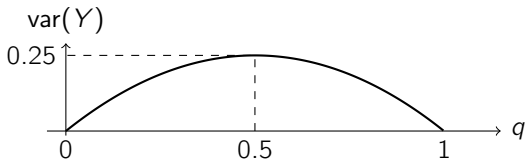
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Let $Y \sim \text{Uniform}(0, L)$. Compute $\text{var}(Y)$ and CV.

The Normal distribution

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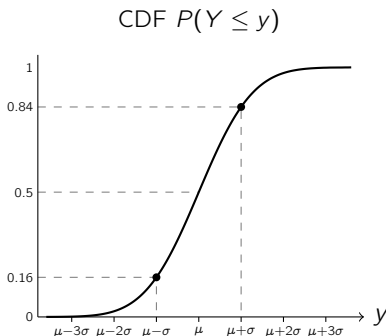
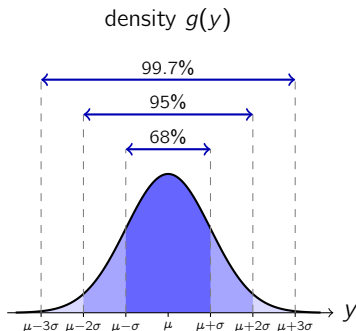
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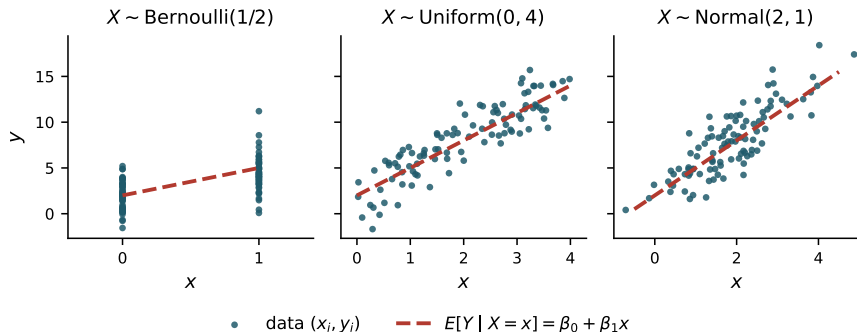
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- X is the predictor
- Y is the response variable
- β_1 is called the regression coefficient of X .
- The variable ϵ may be referred to as an exogenous noise source. Exogeneity means this noise is independent of X .

What data from this model looks like

Data simulated from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\varepsilon^2)$



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- 3) Homoskedasticity: $\text{var}(Y | X = x)$ does not depend on x .

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This formula connects the regression coefficient to the covariance (and hence to expectations we can compute from data).